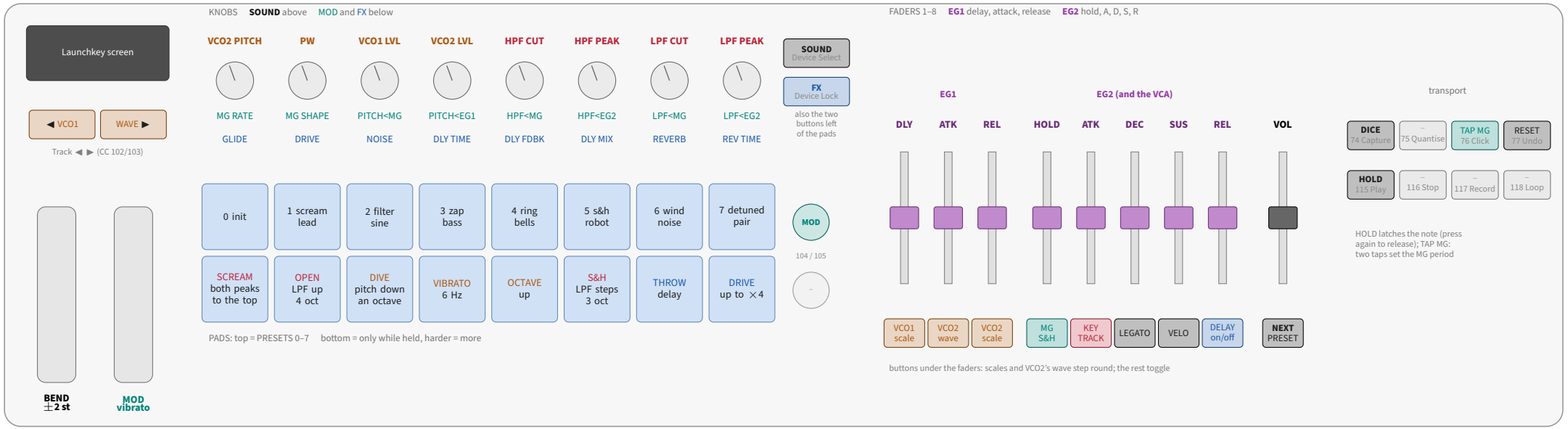


Signal path (as on the MS-20's panel)

VCO1 (triangle, saw, pulse, noise) + VCO2 (saw, square, narrow pulse, ring) + noise → DRIVE → HPF → LPF → VCA (EG2) → delay → reverb → VOL. MG (ramp down – triangle – ramp up, or S&H), EG1 and EG2 modulate pitch and the two cutoffs. **Mono**: the last key sounds; releasing it returns to the key still held.

The filters

Both are 2-pole (12 dB/oct) with clipping in the feedback loop, as the MS-20's Korg-35. PEAK past **91 %** (knob 116): the filter **self-oscillates**, a sine at the cutoff that the clipping keeps in bounds; louder input pushes against it (the scream). HPF then LPF: two peaks make a band-pass with two resonances. KEY TRACK: both cutoffs follow the keys an octave per octave, so a self-oscillating filter plays in tune.

Oscillators (fader buttons 1–3, Track)

VCO1 wave	Track ◀▶: triangle, saw, pulse (width: PW), noise
VCO1 scale	32', 16', 8', 4' (8' = the key's pitch)
VCO2 wave	saw, square, narrow pulse (15 %), ring (VCO1's square × VCO2's square)
VCO2 scale	16', 8', 4', 2'
VCO2 PITCH	±12 st, fine near the centre (centre = in tune)

Envelopes

EG1: after DELAY, rises in ATTACK, stays up while the key is held, falls in RELEASE. PITCH<EG1: the pitch starts that far off (right of centre: above, a zap; left: below, a scoop) and reaches the note as EG1 rises, so held notes stay in tune (the MS-20 adds EG1 to the pitch instead). EG2: attack, decay, sustain, release; HOLD keeps it going that long after the key is up. It drives the VCA and the filters' EG2 amounts. **LEGATO**: keys played over a held key do not restart the envelopes.

Pads

Top = presets 0–7. **Bottom**, while held; strike velocity, then pressure, sets the amount:

SCREAM	both PEAKs towards the top
OPEN	LPF cutoff up to 4 octaves
DIVE	pitch down, up to an octave
VIBRATO / OCTAVE	6 Hz ±0.7 st / an octave up
S&H	random LPF steps at the MG rate, up to 3 oct
THROW	delay throw (echoes ring on)
DRIVE	drive up to ×4

Wheels and transport

MOD wheel: vibrato from the MG, up to 1 st (on top of PITCH<MG). Bend ±2 st. DICE rolls waves, scales and the SOUND and MOD pages (envelopes, FX, volume stay). RESET = preset 0.

Inside

Filters: Sallen–Key 2-pole, $H_{LP} = \omega^2/(s^2 + (2-k)\omega s + \omega^2)$, zero-delay feedback (trapezoidal), a cubic clip in the feedback; each sample solves one scalar equation, which has exactly one root, by 2 Newton steps (measured error $< 6 \cdot 10^{-5}$). Self-oscillation from $k = 2$. The voice runs at 96 kHz (band-limited oscillators, drive and filters) and a half-band filter (-79 dB) brings it to 48 kHz. Envelopes per sample; pitch and cutoffs at 12 kHz. Code: `lib/raw_voice.h`.

Load

DSP 70.5–79.5 % (constant: the voice costs the same with or without a note; most for the ring), patch RAM 22 of 44 KB.

Presets: top pads = 0–7; NEXT PRESET (fader button 9) steps; RESET (Undo) = 0

(chosen from the parameter meanings, not yet tuned by ear)

0 init	One saw through the LPF (peak 0.55 of the way), EG2 opens it 3.8 octaves. The power-on sound.
1 scream lead	Pulse + saw a little detuned, HPF at 180 Hz, LPF peak just below self-oscillation, drive, legato, glide, delay.
2 filter sine	The LPF self-oscillates and tracks the keys (KEY TRACK): a sine played by the filter (about 10 cents flat; LPF CUT steps are coarse), a little triangle and noise to start it.
3 zap bass	Saw + square at 16'; each note starts 13 st above and falls to pitch in 40 ms (EG1); short EG2 into a resonant LPF; velocity.
4 ring bells	VCO2 on ring, a fifth up, short-decay EG2, delay.
5 s&h robot	MG on S&H at 6.5 Hz steps the LPF 2.5 octaves around a resonant cutoff.
6 wind noise	VCO1 on noise; both filters resonant and slowly swept by the MG; slow EG2.
7 detuned pair	Pulse at 8' + saw at 4', 11 cents apart, EG2 into the LPF.

A preset sets the waves, scales, toggles, the three knob pages and the eight envelope faders; the volume stays.

Recipes

Scream: LPF PEAK past 91 %, sweep LPF CUT, then add DRIVE or VCO levels: the input fights the resonance. **Two peaks**: HPF and LPF PEAK both at 3/4, cutoffs a few octaves apart: a hollow, talking band. **Zap**: PITCH<EG1 right of centre, EG1 ATTACK at 1/3 (20 ms), legato off. **Filter as oscillator**: VCO levels down, LPF PEAK full, KEY TRACK on; tune with LPF CUT. **Sample and hold**: MG S&H on, LPF<MG up, MG RATE to taste (or TAP MG).

Where the values fall (knob turned . . .)

	0	1/4	1/2	3/4	full
HPF / LPF CUT (Hz)	20	115	660	3600	20000
PEAK (Q)	0.5	0.7	1.1	2.8	self-osc.
VCO2 PITCH (st)	−12	−3.1	0	+2.9	+12
PW	50 %	39 %	27 %	16 %	5 %
MG RATE (Hz)	0.1	0.48	2.3	11	51
MG SHAPE	ramp down		triangle		ramp up
PITCH<MG (st)	0	0.76	3	6.7	12
PITCH<EG1 (st, start)	−24	−6.2	0	+5.8	+24
HPF/LPF<MG (oct)	0	0.3	1.3	2.8	5
HPF<EG2 / LPF<EG2 (oct)	0	1.5 / 2	3 / 4	4.5 / 6	6 / 8
GLIDE (s)	0	0.13	0.5	1.1	2
DRIVE (gain)	0.5	1	2	4	8
attack, decay, release (s)	0.001	0.01	0.1	1	10
EG1 DELAY, EG2 HOLD (s)	0	0.13	0.5	1.1	2

Something seems wrong?

Silence: VCO levels and NOISE at 0, LPF CUT at the bottom (or HPF CUT at the top), EG2 SUSTAIN 0 with a short decay, or VOL down. **A note that will not stop**: HOLD is latched (Play). **Out of tune**: VCO2 PITCH off centre, or PITCH<EG1 off centre with a long EG1 ATTACK or DELAY (the note arrives late). **Lost**: RESET (Undo). **Another patch** (on the SD card): hold both buttons left of the pads and tap a top pad: 1 freak, 2 grains, 3 groove, 4 soliton, 5 aeolian, 6 raw, 7 quartet.